Strict connection–Dirac spectral dominance for every finite graph

Research proof and Lean formalization
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Abstract

Let L be the simplex-intersection connection matrix of a nonempty finite graph, viewed as its vertex-and-edge simplicial complex, and let D be its signed simplicial Dirac matrix. For every $1 \le k \le v+m$ we prove $S_k(L) - S_k(D) \ge \sqrt{2} - 1 > 0$, where S_k sums the k largest eigenvalues with multiplicity, v is the number of vertices, and m the number of edges. Isolated vertices and disconnected graphs are allowed. This establishes, with a uniform strict gap, the dimension-at-most-one case of Knill's connection–Dirac conjecture. The higher-dimensional conjecture remains outside the result. The proof uses incidence matrices, orthonormal eigenvector lifts, the Ky Fan variational principle, and scalar inequalities.

1 Statement and source matrices

For a real symmetric $N \times N$ matrix A write $\lambda_1(A) \ge \cdots \ge \lambda_N(A)$ and $S_k(A) = \sum_{i=1}^{k} \lambda_i(A)$. Knill [1, Conjecture 1, §5.3] asks whether the connection matrix weakly dominates the Dirac matrix in this sense for every finite abstract simplicial complex. We prove the full graph case.

Let G be a finite simple graph with $v \ge 1$ vertices and m edges. Its nonempty simplices are precisely its vertices and edges; no larger clique is filled. Thus a triangle means its boundary complex. For simplices s, t put $L_{st} = 1$ if $s \cap t \neq \varnothing$, and 0 otherwise. Let d be an oriented simplicial boundary matrix and $D = d + d^{\mathsf{T}}$. Changing edge orientations conjugates D by a diagonal orthogonal matrix.

Theorem 1. *For every such graph and every* $1 \le k \le v+m$,

$$S_k(L) - S_k(D) \ge \sqrt{2} - 1 > 0.$$

The constant is a uniform lower bound; no optimality is asserted.

Order vertices before edges. Let U be unsigned vertex–edge incidence, and C signed incidence, with one $+1$ and one -1 in each column. Write δ_i for the vertex degrees, $\Delta = \operatorname{diag}(\delta_i)$, $Q = UU^{\mathsf{T}}$, and $H = CC^{\mathsf{T}}$. The literal matrices are

$$L = \begin{pmatrix} I & U \\ U^{\mathsf{T}} & U^{\mathsf{T}}U - I \end{pmatrix}, \qquad D = \begin{pmatrix} 0 & C \\ C^{\mathsf{T}} & 0 \end{pmatrix}. \tag{1}$$

In the edge block, distinct simple edges share at most one vertex and $(U^{\mathsf{T}}U)_{ee} = 2$. Consequently (1) is exactly the intersection definition. Also

$$Q_{ii} = \delta_i, \qquad \operatorname{tr} Q = \operatorname{tr} H = 2m, \qquad H \preceq 2\Delta, \tag{2}$$

the last inequality following from $(x_i - x_j)^2 \le 2x_i^2 + 2x_j^2$ on every edge. Here $\preceq$ denotes ordinary positive-semidefinite order. Let T_k be the sum of the k largest degrees, for $k \le v$.

2 The connection eigenvector lift

Set

$$\phi(q) = \frac{q + \sqrt{q^2+4}}{2}, \qquad e(q) = \frac{\sqrt{q^2+4}}{2} - 1.$$

For $q \ge 0$ one has $\phi(q) > 0$, $\phi(q)^2 - q\phi(q) = 1$, and $\phi(q) = 1 + q/2 + e(q)$ with $e(q) \ge 0$. If $Qu = qu$, direct substitution into (1) shows that

$$\big((\phi(q)+1)u,\ \phi(q)U^{\mathsf{T}}u\big) \tag{3}$$

is an L-eigenvector with eigenvalue $\phi(q)$. This formula also works at $q = 0$. It is nonzero for $u \neq 0$. An orthonormal Q-eigenbasis yields mutually orthogonal vectors (3), because $\langle U^{\mathsf{T}}u_i, U^{\mathsf{T}}u_j\rangle = \langle u_i, Qu_j\rangle = q_j\delta_{ij}$. Normalize these vectors and apply the Ky Fan principle. If $q_1 \ge \cdots \ge q_v \ge 0$ are the eigenvalues of Q, then

$$S_k(L) \ge \sum_{i=1}^{k} \phi(q_i) \quad (1 \le k \le v). \tag{4}$$

The incidence reduction is consistent with Knill's earlier hydrogen identity [2, Theorem 1]; the block reduction itself is a known ingredient, not an additional novelty claim.

The diagonal Ky Fan inequality gives $\sum_{i=1}^{k} q_i \geq T_k$. If $m > 0$, choose an edge $\{a, b\}$ and $z = e_a + e_b$. Then $\|z\|^2 = 2$ and $z^{\mathsf{T}} Q z = \delta_a + \delta_b + 2 \geq 4$, so $q_1 \geq 2$. Since e increases on $[0, \infty)$,

$$S_k(L) \geq k + T_k/2 + e(q_1) \geq k + T_k/2 + \sqrt{2} - 1 \qquad (1 \leq k \leq v). \tag{5}$$

3 Upper bounds for the signed Dirac matrix

Consider any orthonormal family of positive D-eigenvectors $w_i = (a_i, b_i)$ with eigenvalues $\mu_i > 0$. The block equations are $Cb_i = \mu_i a_i$ and $C^{\mathsf{T}} a_i = \mu_i b_i$. Adjointness gives

$$\mu_i \langle b_i, b_j \rangle = \mu_j \langle a_i, a_j \rangle, \qquad \mu_j \langle b_i, b_j \rangle = \mu_i \langle a_i, a_j \rangle.$$

Adding and using $\mu_i + \mu_j > 0$ shows that the two inner products are equal. Their sum is δ_{ij}, hence $u_i = \sqrt{2} a_i$ is an orthonormal family in $\mathbb{R}^v$ and $H u_i = \mu_i^2 u_i$. This treats repeated eigenvalues directly.

Take the positive eigenvalues among the first k eigenvalues of D, and denote their number by r. Dropping the nonpositive entries increases the sum. For $k \leq v$ we have $r \leq k$; using (2), Ky Fan, and nonnegativity of the degrees gives

$$\sum_{i=1}^{r} \mu_i^2 = \sum_{i=1}^{r} \langle u_i, H u_i \rangle \leq 2T_r \leq 2T_k.$$

Cauchy–Schwarz therefore proves

$$S_k(D) \leq \sum_{i=1}^{r} \mu_i \leq \sqrt{r \sum_{i=1}^{r} \mu_i^2} \leq \sqrt{2kT_k}. \tag{6}$$

For arbitrary $k \leq v + m$, orthonormality instead gives $r \leq v$ and $\sum_{i=1}^{r} \mu_i^2 \leq \operatorname{tr} H = 2m$. The scalar inequality $t \leq (t^2 + 1)/2$ for $t \geq 0$ yields

$$S_k(D) \leq \frac{\sum_{i=1}^{r} \mu_i^2 + r}{2} \leq m + v/2. \tag{7}$$

This argument only needs upper bounds; it makes no assumption about the zero multiplicity in the full Dirac spectrum.

4 Completion of the comparison

Suppose first $m > 0$ and $k \leq v$. Since

$$(k + T_k/2)^2 - 2kT_k = (k - T_k/2)^2 \geq 0,$$

we have $k + T_k/2 \geq \sqrt{2kT_k}$. Subtract (6) from (5) to obtain the claimed gap.

For the remaining indices we use an alternative to a full inertia calculation. The quadratic identity, for $x = (a, b)$,

$$x^{\mathsf{T}}(L + I)x = \|a\|^2 + \|a + Ub\|^2 \geq 0$$

proves $\lambda_i(L) \geq -1$. Put $g = \sum_{i=1}^{v} \phi(q_i)^{-1} > 0$. The identity $\phi(q) - \phi(q)^{-1} = q$ and (2) give $\sum_i \phi(q_i) = 2m + g$. By Cauchy–Schwarz,

$$v^2 \leq \Big(\sum_i \phi(q_i)\Big)\Big(\sum_i \phi(q_i)^{-1}\Big) = (2m + g)g.$$

Simplicity gives $2m \leq v(v - 1)$, and $v \geq 1$ implies $2m + 1 \leq v^2$. If $g < 1$, then $(2m + g)g < 2m + 1$, a contradiction. Thus $g \geq 1$ and (4) gives $S_v(L) \geq 2m + 1$. For $v \leq k \leq v + m$ it follows that

$$S_k(L) \geq 2m + v + 1 - k.$$

On the other hand, the diagonal of L consists of ones, so Ky Fan gives $S_k(L) \geq k$. Averaging these two lower bounds yields

$$S_k(L) \geq m + v/2 + 1/2.$$

Together with (7), this proves a gap at least $1/2 > \sqrt{2} - 1$ for these indices. Finally, if $m = 0$, then $L = I_v$ and $D = 0$, so the gap is $k \geq 1$. All cases of Theorem 1 follow.

5 Formal development and scope

The accompanying seventeen Lean modules contain 211 declarations and a closed literal source endpoint. In addition to the incidence identities, signed spectral variational principle, normalized connection eigenvector family, and both connection lower bounds, the proof constructs the normalized positive Dirac eigenvectors and proves both required Dirac upper bounds. The endpoint `source_graph_strict_gap` quantifies over every nonempty finite family of nonempty faces of size one or two, closed under nonempty subsets, and all $1 \leq k \leq |G|$. Its matrices are the actual face-intersection matrix and the signed face-boundary matrix plus its transpose. Reindexing, including removal of unused ambient vertices, is proved to preserve both spectral sums. No rank, singular-value, or eigenvector-existence assertions are left as premises of the endpoint. The replay uses Lean 4.34.1 and mathlib revision `d13f23b7`, treats warnings as errors, and checks every declaration's axioms. Only `propext`, `Classical.choice`, and `Quot.sound` occur.

The contribution is one general comparison for every graph, rather than separate claims for its helpers, special cases, or constant. The source problem for complexes with higher-dimensional faces is untouched. Knill's hydrogen identity, incidence formulas, spectral variational facts, and scalar inequalities are credited as established ingredients. The accompanying records preserve the primary sources and a bounded prior-art investigation; no worldwide priority claim or complete literature census is asserted.

References

[1] O. Knill, *Remarks about connection and Dirac matrices*, arXiv:2601.18071v2 (29 July 2026), https://arxiv.org/abs/2601.18071.

[2] O. Knill, *The hydrogen identity for Laplacians*, arXiv:1803.01464v1 (2018), https://arxiv.org/abs/1803.01464.